Modelling Tectonic Plate Movement

Aim: To model the Earth's tectonic plate movement.

Equipment:

- Safety goggles
- Large 1l beaker
- 300ml treacle
- Biscuit (halved)
- Tripod
- Gauze
- Bunsen burner
- Heat proof mat

Safety Information

Hot syrup – risk of burns, may spit when boiling.

Wear goggles at all times. If burnt, put area under running cold water for 10 minutes and inform teacher immediately.

Glassware – risk of cuts if broken.

Apply pressure to wounds and inform teacher immediately.

Bunsen – risk of burns.

Tie back hair, handle with caution.

If burnt, put area under running cold water for 10 minutes and inform teacher immediately.

Method:

Step 1: Set up the apparatus as shown in the diagram

Step 2: Pour the treacle into the beaker

Step 3: Lay the two halves of the biscuit onto the treacle, with a small gap between them

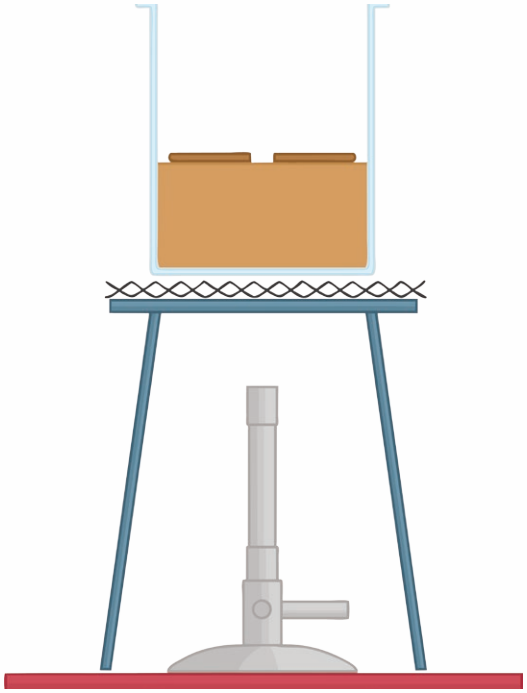
Step 4: Light your Bunsen burner on a safety flame (hole closed)

Step 5: Move the Bunsen burner underneath the tripod (remember: to safely move a Bunsen burner while it is lit, hold it where the rubber tubing connects to the base)

Step 6: Gently heat the treacle with a blue flame until it begins to bubble

Step 7: Observe the biscuits carefully, do not let the treacle boil too much

Step 8: Remove the heat to control the bubbling of the treacle



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In 1912, meteorologist Alfred Wegener describe his theory of continental drift, and the idea that the different continents would have fit together like a jigsaw puzzle. He couldn't explain how though, and it was about 50 years later that the theory of plate tectonics arose.

Questions:

1. What does the treacle represent in this demonstration?
2. What do the biscuit halves represent in this demonstration?
3. How does the heated treacle cause the biscuits to move?
4. Use the model to describe how earthquakes occur.

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Answers:

1. What does the treacle represent in this demonstration?
   The molten magma layer of the earth

2. What do the biscuit halves represent in this demonstration?
   The tectonic plates which make up the earth's crust

3. How does the heated treacle cause the biscuits to move?
   As the treacle is heated from underneath, the particles gain more energy and begin to move faster and further apart, making the area of treacle less dense. The less dense area of treacle floats upwards and into the underside of the biscuits. As more treacle rises up the top layer of treacle is forced to move sideways and with it carries the biscuit. The heat is being transferred by convection.

4. Use the model to describe how earthquakes occur?
   As the tectonic plates pass one another they rub at their boundary and cause friction. When the forces become unbalanced then the plates will jerk past one another, causing an earthquake.